

Raidel Almeida

(786) 399-8029 | contact@raidel.dev | linkedin.com/in/raidel-almeida | github.com/raidel-a | raidel.dev

EDUCATION

Florida International University
Bachelor of Science in Computer Science

Miami, FL
Dec 2026

EXPERIENCE

Miami Heat / 601 Analytics

Miami, FL

Data Analytics/Machine Learning Intern

May – July 2025

- Developed a **personalized shopping recommendation system** to improve Miami Heat's online retail conversion rate using advanced machine learning techniques
- Implemented **two distinct ML models**: Apriori-Rule Based Mining (ARM) achieving **36.5% F1 score** and a hybrid Content-Based/Collaborative Filtering model with **60.2% accuracy**
- Processed and analyzed **5 datasets containing 600,000+rows** spanning **8 years** of customer data using **Python, Pandas, and Dask** for efficient data manipulation and feature engineering
- Engineered hybrid recommendation system combining **TF-IDF, cosine similarity, and PyTorch neural networks** with **RFM customer segmentation** and comprehensive data visualizations using **Plotly Express** and **Seaborn**

LEADERSHIP

INIT FIU Executive Board

Director of Technology

Jan – Dec 2025

- Spearheaded development of **ShellHacks web application** to handle **2,000 +applications** and **1,300+participants** for Florida's largest hackathon, implementing **robust architecture** to ensure **zero downtime** during peak usage
- Implement comprehensive system featuring **specialized dashboards** for hackers, administrators, and volunteers, along with integrated **bulk email service** and **QR code scanning** functionality for event check-ins
- Complete migration from Next.js to Astro, Preact, TursoDB (SQL/SQLite) and Go tech stack to enhance application performance, maintainability, and simplify future development cycles

Explore Hardware Technical Lead

Jan – Dec 2024

- Taught **12+technical workshops** reaching **500+students**, including robotics, Arduino programming, and 3D modeling workshops with average of **30+students**
- Crafted **detailed documentation and materials** for workshops, enabling continued learning beyond sessions
- Established **safety protocols and standards** for hardware lab operations, ensuring compliance with guidelines
- Organized Florida's **largest hackathon, ShellHacks**, attracting **1400+participants** and **30+industry sponsors**

PROJECTS

🔗 butterfly-IOT Garden | C++, React, ESP32, Raspberry Pi, InfluxDB, Three.js

Aug – Dec 2024

- Led a team of 10+members** to develop a cost-effective IoT system monitoring environmental data (soil moisture, temperature, and more) for FIU's GCI Garden, providing the public and gardeners with **real-time analytics**
- Designed a **distributed sensor network** using **ESP32 nodes** paired with a **Raspberry Pi hub** running **Mosquitto MQTT, Node-RED workflows, and InfluxDB** for time-series data storage
- Engineered a **public visualization portal** with **React** and **Three.js**, featuring interactive charts (**ReCharts**), while maintaining **internal dashboards via Grafana** for analytics
- Implemented **Dockerized infrastructure** on Raspberry Pi for seamless deployment of **MQTT broker, database, and visualization services**, ensuring **scalable data processing** for 50+concurrent sensor streams

🔗 Mimir | Go, Next.js, React, TypeScript, Python, WeaviateDB

Knighthacks 2024

- Won ServiceNow Challenge (**1st place** at UCF's Hackathon) by developing an AI-powered documentation platform that streamlines technical knowledge management
- Implemented a scalable backend in **Go** with **WeaviateDB** vector database, implementing **real-time clustering** for intelligent document categorization
- Architected responsive frontend using **Next.js** and **TypeScript**, featuring dynamic search, chat interface, and automated documentation suggestions
- Integrated **OpenAI's GPT-4** for intelligent ticket clustering and automated documentation generation